Final Project Report

Project 1: Network Mapper

Student Name	**Steve**
Project	Project 1 — Network Mapper
Specialization	Network Scanning and Enumeration
Scanner Machine	Kali Linux (hostname: V2310) — 192.168.10.6
Target Subnet	192.168.10.0/24
Tool Used	Nmap

1. Objective

Discover everything on the lab network and document it like an asset inventory: run a ping sweep to find live hosts on the subnet, port-scan every live host, run service and version detection on the open ports, and present the results as a network map and an inventory table (host, ports, services, versions).

2. Scope and Lab Environment

All scanning was performed from Kali Linux running in VMware, against the lab subnet 192.168.10.0/24. The two VMware virtual machines identified on the network (MAC vendor: VMware) are the Windows VM at 192.168.10.33 and the Windows Server VM (V2434 / STEVE) at 192.168.10.71, which were scanned in full (port scan plus service and OS detection). One further discovered host, the mobile phone REDMI-15C at 192.168.10.49, was port-scanned and returned all ports filtered.

3. Tools Used

Tool	Purpose
Kali Linux (VMware)	Attacker / scanner machine used to run all scans
Nmap	Host discovery (ping sweep), port scanning, service/version detection and OS detection
Terminal output files (.txt)	Each scan was saved with the -oN option (discovery.txt, ports-102.txt, services-102.txt)

4. Procedure

Step 1 — Host discovery (ping sweep) on the full subnet, saved to discovery.txt:

```
nmap -sn 192.168.10.49/24 -oN discovery.txt
```



Figure 1 — Step 1: Ping sweep of 192.168.10.0/24, hosts appearing (saved to discovery.txt).

What this screenshot shows: The ping sweep result. The command used -sn, which means ping scan only: Nmap asked every address in the subnet who is alive, without scanning ports yet. Each block is one live host: its name (for example Broadcom.Home, KaliLinux, OPPO-F7), its

IP address, how fast it replied (latency), and its MAC address with the manufacturer (vendor) of its network card. This is the raw material for the host discovery table in Section 5. The Kali machine itself appears in the list at 192.168.10.6.

Step 2 — Port scan of the discovered mobile host (192.168.10.49):

```
nmap 192.168.10.49 -oN ports-102.txt
```

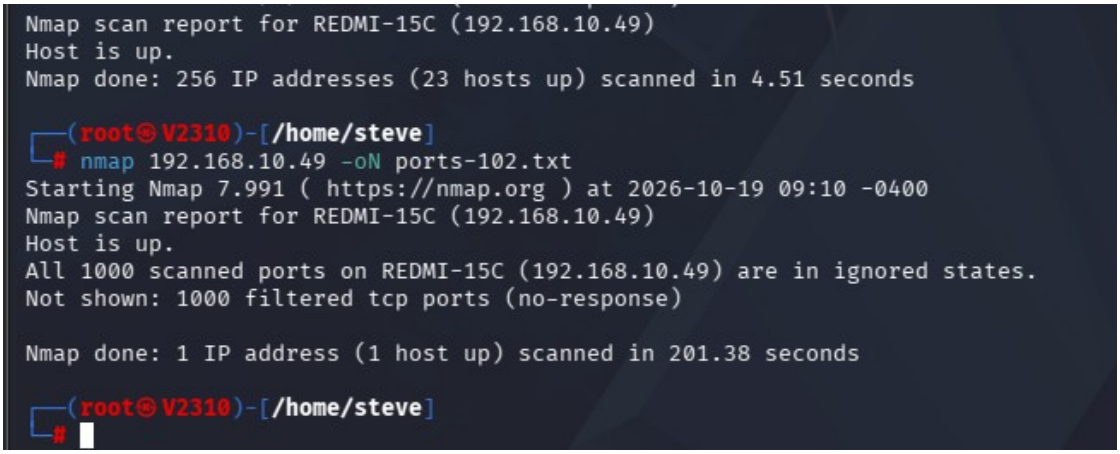


Figure 2 — Step 2: End of the discovery scan (256 addresses, 23 hosts up), then the port scan of REDMI-15C (192.168.10.49) — all 1000 ports filtered.

What this screenshot shows: Two results in one screenshot. At the top, the discovery scan finishes: 256 IP addresses were scanned and 23 hosts were up, in only 4.51 seconds. Below it, the port scan of REDMI-15C (192.168.10.49) shows the opposite: it took 201.38 seconds and every one of the 1000 scanned ports was filtered (no-response). Filtered means the phone silently dropped the probes instead of answering, so no open services could be found on it. Phones are normally well protected like this.

Step 3 — Service, version and OS detection on the Windows VM (192.168.10.33):

```
sudo nmap -sV -O 192.168.10.33 -oN services-102.txt
```

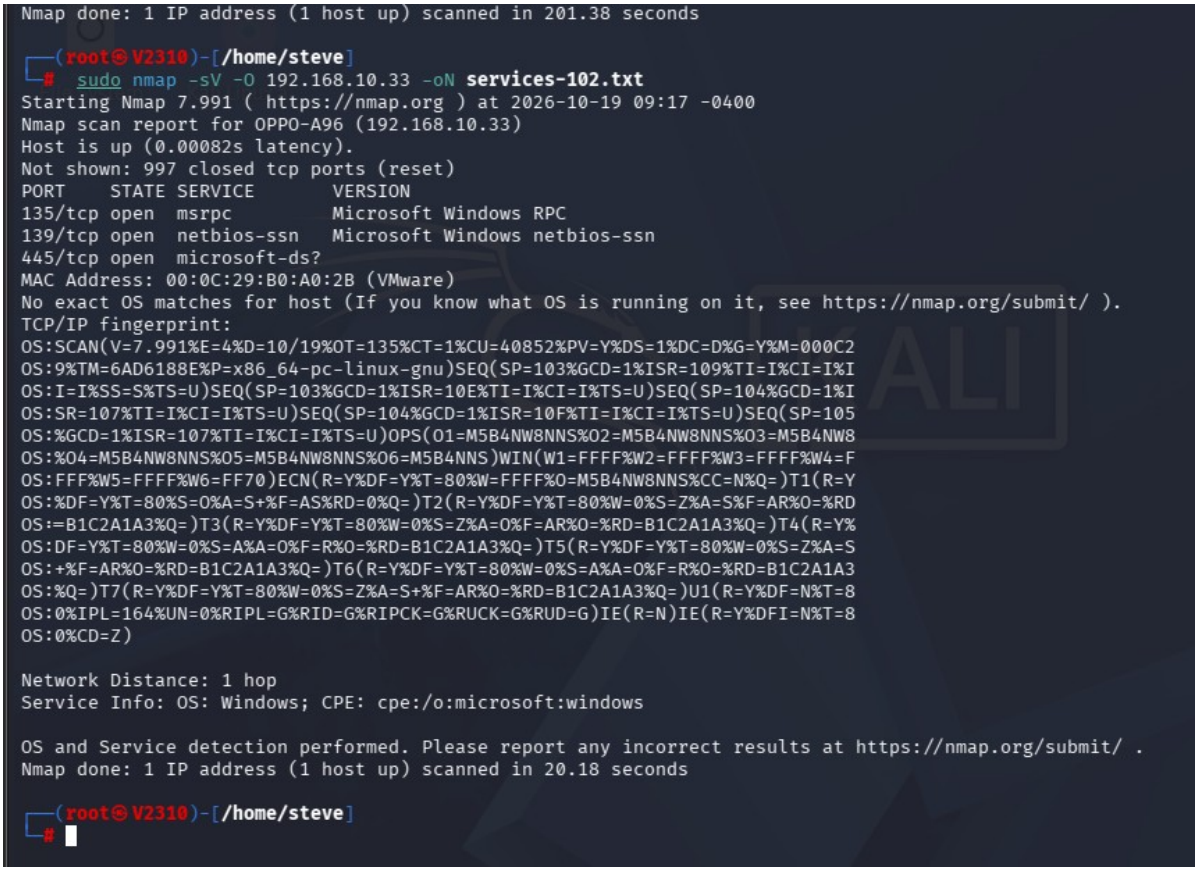


Figure 3 — Step 3: Service and OS detection on 192.168.10.33 (OPPO-A96, Windows VM) — ports 135, 139 and 445 open.

What this screenshot shows: Service and OS detection on the Windows VM at 192.168.10.33. The -sV switch asked each open port what is really running on it: Microsoft Windows RPC on 135, NetBIOS on 139 and SMB (microsoft-ds, Windows file sharing) on 445, while 997 other ports were closed. The MAC address starting 00:0C:29 proves this machine is a VMware virtual machine. Nmap could not match an exact OS fingerprint, but its Service Info line still identified the machine as Windows.

Step 4 — Port scan of the Windows Server VM (192.168.10.71):

```
nmap 192.168.10.71 -oN ports-102.txt
```

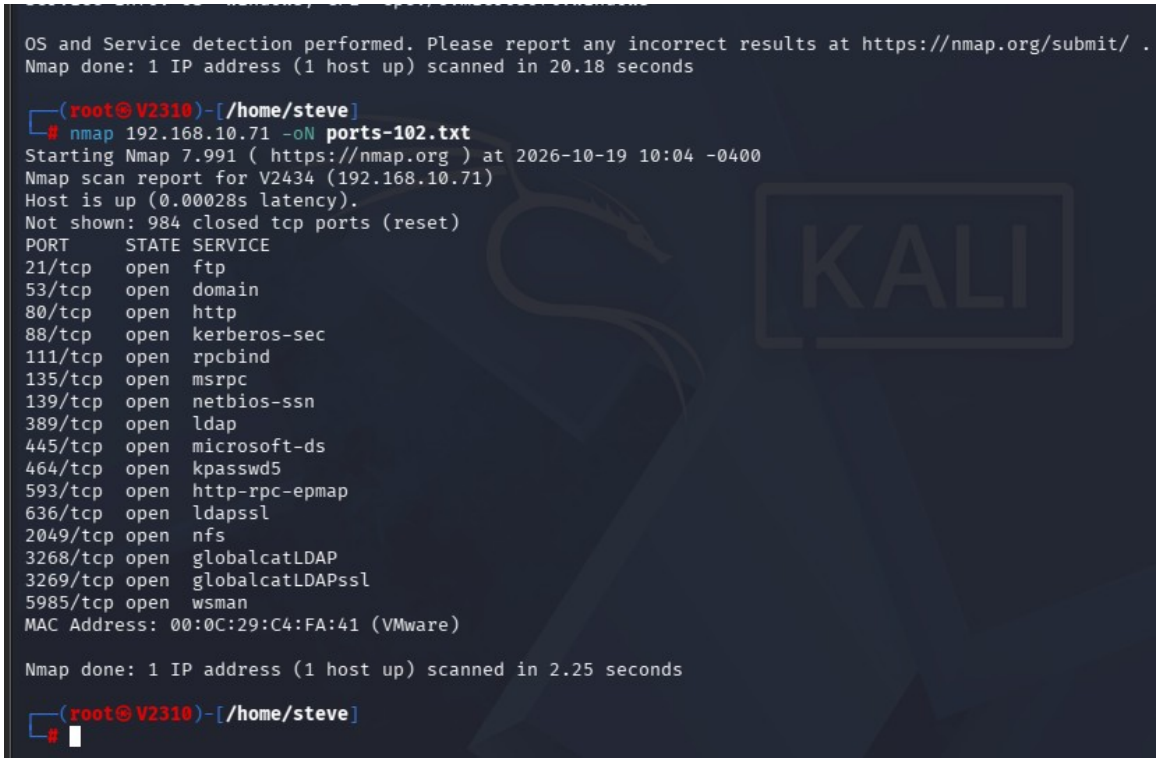


Figure 4 — Step 4: Port scan of 192.168.10.71 (V2434) — 16 open ports.

What this screenshot shows: The port scan of the Windows Server VM at 192.168.10.71 (V2434). 16 ports are open and 984 are closed. The service names in this list (ftp, domain, http, ldap and so on) are the standard services normally found on those port numbers; they are Nmap's first identification from the port number alone. The next step checks what is really running on each port, which is why both scans are needed.

Step 5 — Service, version and OS detection on the Windows Server VM (192.168.10.71):

```
sudo nmap -sV -O 192.168.10.71 -oN services-102.txt
```

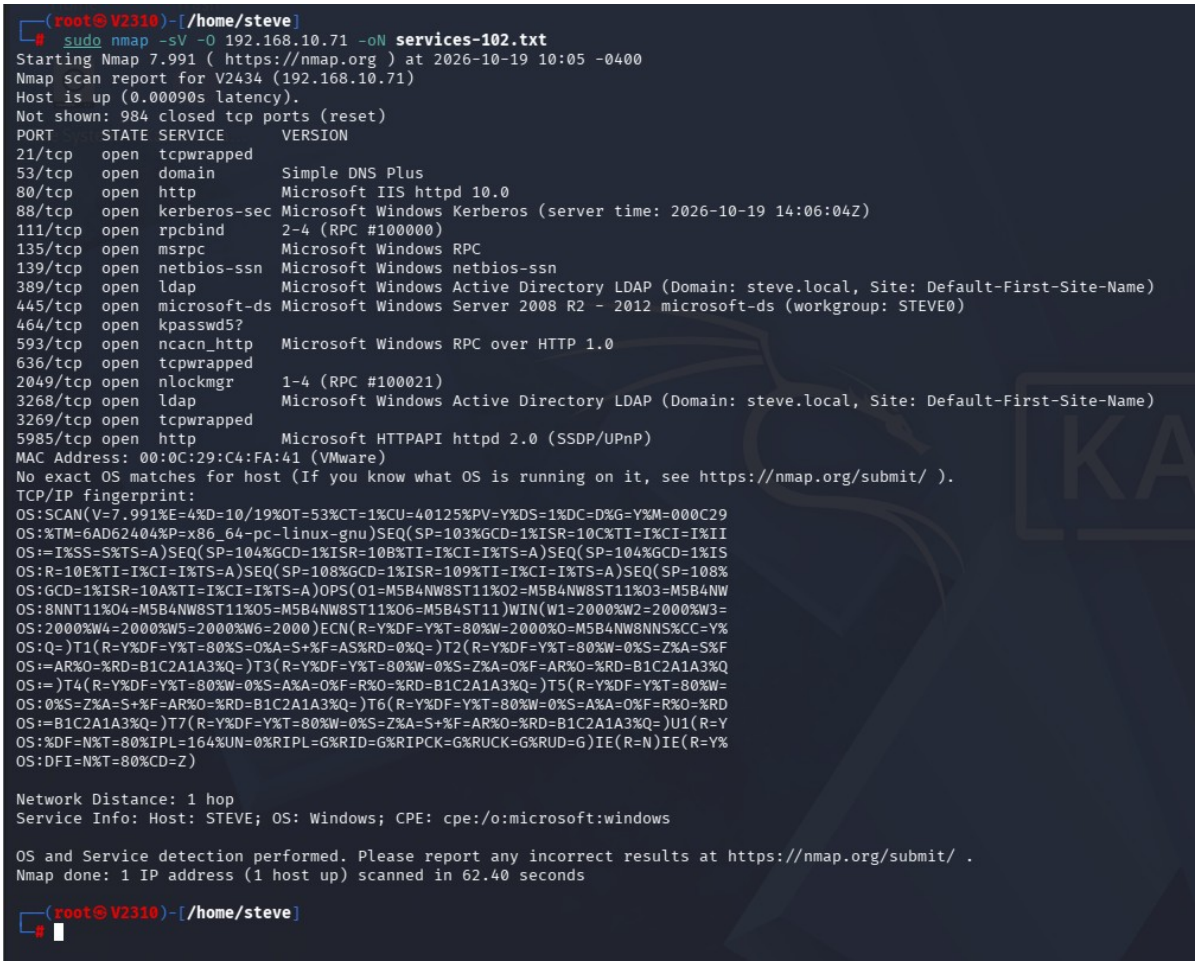


Figure 5 — Step 5: Service and OS detection on 192.168.10.71 — Active Directory LDAP (steve.local), IIS 10.0, Kerberos, DNS.

What this screenshot shows: The most important result of the project. Version detection on 192.168.10.71 returned real software names and versions: Microsoft IIS 10.0 web server on port 80, Simple DNS Plus on 53, and Active Directory LDAP on 389 and 3268, which even announced the domain name steve.local. Together with Kerberos on 88, this proves the machine is a Windows domain controller. A few ports answered tcpwrapped, meaning the service accepted the connection but sent no banner, so no version could be read. The scan took 62.40 seconds.

5. Host Discovery Results

The ping sweep scanned 256 IP addresses and reported 23 hosts up in 4.51 seconds. The table below is transcribed exactly from the submitted scan screenshots (Figures 1 and 2, at Steps 1 and 2 of the Procedure). Hosts whose details are not readable in the screenshots are marked with a dash.

#	Hostname	IP Address	MAC Address	Vendor / Notes
1	Broadcom.Home	192.168.10.1	74:12:BB:E2:67:F2	Fiberhome Telecommunication Technologies

				(gateway)
2	—	192.168.10.5	A0:59:50:A8:62:81	Intel Corporate
3	KaliLinux	192.168.10.6	94:E2:3C:6C:38:8D	Intel Corporate (scanner machine)
4	OPPO-F7	192.168.10.10	56:FE:D4:41:63:C1	Unknown
5	Mirza-s	192.168.10.11	34:CF:F6:7C:F5:E7	Intel Corporate
6	PNY-ARFA-AP1	192.168.10.13	70:81:05:42:6F:E8	Cisco Systems (access point)
7	iPhone	192.168.10.16	92:0B:60:77:46:23	Unknown
8	DESKTOP-6DAJUGO	192.168.10.17	B0:3C:DC:36:2A:59	Intel Corporate
9	PNYF8	192.168.10.20	9C:14:63:9E:02:19	Zhejiang Dahua Technology
10	OPPO-A96	192.168.10.33	00:0C:29:B0:A0:2B	VMware — Windows VM, scanned in full (Section 6)
11	zainab	192.168.10.35	B2:10:33:5B:7E:24	Unknown
12	DESKTOP-1JPJLOS	192.168.10.36	98:09:CF:1F:ED:22	OnePlus Technology (Shenzhen)
13	Z2453	192.168.10.41	2A:2B:E5:A7:54:54	Unknown
14	A70	192.168.10.46	26:14:11:BC:F4:1D	Unknown
15	Tayyaba-s-A06	192.168.10.47	06:0B:30:4A:33:3A	Unknown
16	vivo-1906	192.168.10.48	BE:7F:78:08:1D:2B	Unknown
17	REDMI-15C	192.168.10.49	—	Mobile phone, port-scanned (Section 6)
18	PC	192.168.10.51	—	—
19	V2434	192.168.10.71	00:0C:29:C4:FA:41	VMware — Windows Server VM, scanned in full (Section 6)

6. Inventory Table — Ports, Services and Versions

Built from the port scans and the service/version detection scans (Figures 2 to 5, at Steps 2 to 5 of the Procedure). This is the main deliverable of the project.

Host 192.168.10.71 — V2434 / STEVE (Windows Server, VMware). 16 open ports; 984 closed. Nmap OS detection reported: OS Windows (Service Info: Host STEVE). The open LDAP, Kerberos and DNS services together identify this machine as an Active Directory Domain Controller for the domain steve.local.

Host IP	OS	Port	Service	Version
192.168.10.71	Windows	21/tcp	tcpwrapped	—
192.168.10.71	Windows	53/tcp	domain	Simple DNS Plus
192.168.10.71	Windows	80/tcp	http	Microsoft IIS httpd 10.0
192.168.10.71	Windows	88/tcp	kerberos-sec	Microsoft Windows Kerberos
192.168.10.71	Windows	111/tcp	rpcbind	2-4 (RPC #100000)
192.168.10.71	Windows	135/tcp	msrpc	Microsoft Windows RPC
192.168.10.71	Windows	139/tcp	netbios-ssn	Microsoft Windows netbios-ssn
192.168.10.71	Windows	389/tcp	ldap	Microsoft Windows Active Directory LDAP (Domain: steve.local)
192.168.10.71	Windows	445/tcp	microsoft-ds	Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
192.168.10.71	Windows	464/tcp	kpasswd5	—
192.168.10.71	Windows	593/tcp	ncacn_http	Microsoft Windows RPC over HTTP 1.0
192.168.10.71	Windows	636/tcp	tcpwrapped	— (ldaps in port scan)
192.168.10.71	Windows	2049/tcp	nlockmgr	1-4 (RPC #100021)
192.168.10.71	Windows	3268/tcp	ldap	Microsoft Windows Active Directory LDAP (Domain: steve.local)

192.168.10.71	Windows	3269/tcp	tcpwrapped	— (globalcatLDAPssl in port scan)
192.168.10.71	Windows	5985/tcp	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP) — WinRM port

Host 192.168.10.33 — OPPO-A96 (Windows VM, VMware). 3 open ports; 997 closed. Nmap OS detection: no exact OS match, Service Info reported OS: Windows.

Host IP	OS	Port	Service	Version
192.168.10.33	Windows	135/tcp	msrpc	Microsoft Windows RPC
192.168.10.33	Windows	139/tcp	netbios-ssn	Microsoft Windows netbios-ssn
192.168.10.33	Windows	445/tcp	microsoft-ds	—

Host 192.168.10.49 — REDMI-15C (mobile phone). The port scan ran for 201.38 seconds and found no open ports: all 1000 scanned TCP ports were in a filtered (no-response) state, which means the device silently drops scan traffic.

Host IP	OS	Port	Service	Version
192.168.10.49	— (mobile)	All 1000 TCP	filtered (no-response)	No open ports found

7. Network Map

The map below was drawn from the discovery and inventory results above. The gateway (Broadcom.Home, 192.168.10.1) sits at the centre of the 192.168.10.0/24 subnet; the scanner (Kali Linux), the two fully scanned VMware machines, the port-scanned mobile phone, and the remaining discovered hosts are grouped around it.

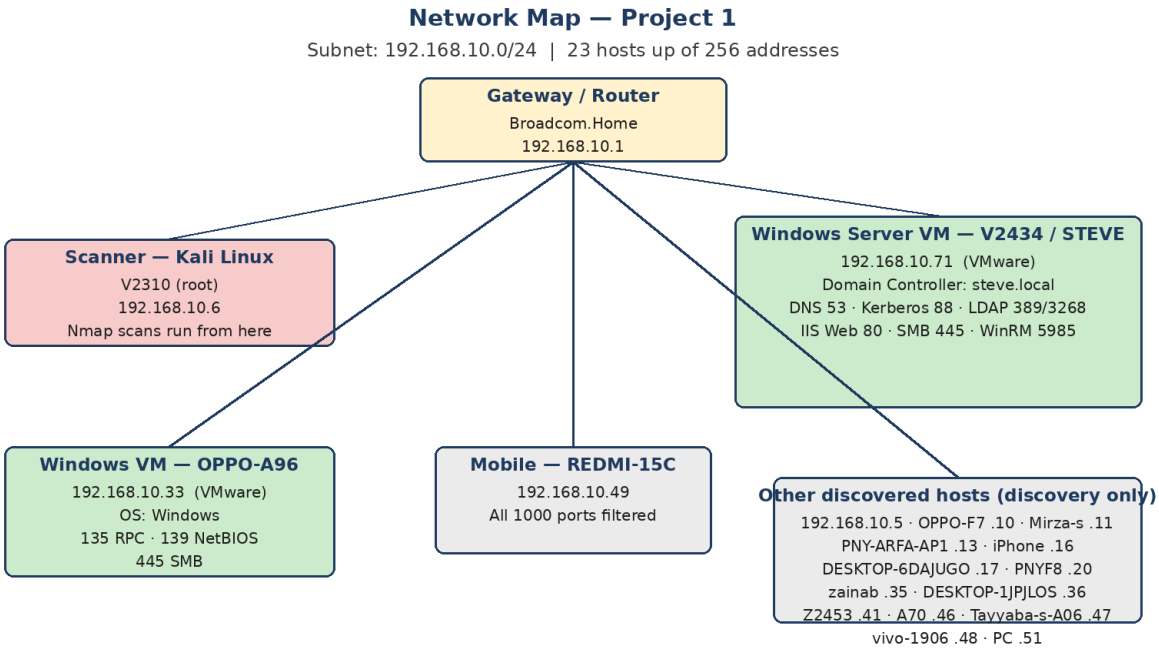


Figure: Network map of 192.168.10.0/24, drawn from the Nmap results in this report.

What this screenshot shows: How to read this map: the gateway/router (192.168.10.1) is at the centre because every device connects through it. The red box is the Kali scanner machine the scans were run from. The two green boxes are the virtual machines that were scanned in full, with their most important services listed. The grey boxes are the phone that was port-scanned and a summary of the other discovered hosts, which were found by the ping sweep but not scanned in detail.

8. Observations

- Host discovery: the ping sweep found 23 live hosts out of 256 addresses on 192.168.10.0/24, including the gateway, the Kali scanner itself, desktop PCs, mobile phones, an access point and two VMware virtual machines.
- Most important finding: 192.168.10.71 is a Windows Active Directory Domain Controller for the domain steve.local — proven by Kerberos (88), DNS (53), LDAP (389 and 3268) running together, and the Active Directory banner returned by the LDAP service.
- The same server also runs a web server (Microsoft IIS 10.0 on port 80), file sharing (SMB on 445) and remote management (WinRM port 5985), so it holds several roles at once.
- The Windows VM at 192.168.10.33 exposes only the classic Windows trio: RPC (135), NetBIOS (139) and SMB (445).
- The mobile phone REDMI-15C (192.168.10.49) returned every scanned port as filtered — a phone's firewall silently dropping the probes — so no services could be identified on it.

- Service identification note: on a few ports of 192.168.10.71 the version scan reported tcpwrapped (21, 636, 3269), meaning the service accepted the connection but returned no banner; those ports are identified by their port-scan service names in brackets in the inventory table.

9. Result

Project 1 was completed against its marking criteria. Host discovery (35): the full subnet was swept and 23 live hosts were recorded with IP, hostname, MAC address and vendor. Service identification (35): every open port on the two VMware lab machines (19 ports in total) was identified with service name and version where the service provided a banner. Documentation (30): results are presented in this report as a host discovery table, a full inventory table and a network map, with the original scan screenshots placed at each step of the Procedure.

— End of Report —